(DBT) ASSIGNMENT-10

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W.

```
root DELIMITER $$
root
rootCREATE PROCEDURE hm_measurse(l INT , w INT)
    -> BEGIN
    -> DECLARE v_perimeter INT;
    -> DECLARE v_area INT;
    -> SET v_perimeter = 2 * (l+w);
    -> SET v_area = l*w;
    ->
    -> SELECT v_perimeter AS Perimeter_of_Rectangle, v_area AS Area_of_Rectangle;
    ->
    -> END
    -> $$
Query OK, 0 rows affected (0.01 sec)

rootDELIMITER ;
root
rootCALL hm_measurse(10 , 20);
+------------------------+-------------------+
| Perimeter_of_Rectangle | Area_of_Rectangle |
+------------------------+-------------------+
|                     60 |               200 |
+------------------------+-------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
root DELIMITER $$
rootCREATE PROCEDURE hm_sqr_cube()
    -> BEGIN
    -> DECLARE v_num INT DEFAULT 7;
    -> DECLARE v_sqr INT;
    -> DECLARE v_cube INT;
    ->
    -> CREATE TABLE temp(
    ->  varable INT,
    ->  squares INT ,
    ->  cubes INT
    -> );
    ->
    -> SET v_sqr = v_num * v_num;
    -> SET v_cube = v_num * v_num * v_num;
    ->
    -> INSERT INTO temp() VALUES(v_num, v_sqr, v_cube);
    -> END
    -> $$
Query OK, 0 rows affected (0.00 sec)

root DELIMITER ;
```

```
root  CALL hm_sqr_cube();
Query OK, 1 row affected (0.02 sec)

root SELECT * FROM temp;
+---------+---------+-------+
| varable | squares | cubes |
+---------+---------+-------+
|       7 |      49 |   343 |
+---------+---------+-------+
1 row in set (0.00 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are C= (F-32) *5/9 F= 9/5*C + 32

```
root DELIMITER $$
rootCREATE PROCEDURE hm_temp_convertor( IN v_temp DOUBLE ,IN v_scale VARCHAR(10), OUT res DECIMAL(10,2) )
    -> BEGIN
    ->
    -> IF v_scale = 'F' THEN
    ->     SET res = (v_temp - 32) *5/9;
    -> ELSEIF v_scale = 'C' THEN
    ->     SET res = 9/5* v_temp + 32;
    ->
    -> ELSE
    ->  SET res = null;
    -> END IF;
    ->
    -> END
    -> $$
Query OK, 0 rows affected (0.00 sec)
```

```
root DELIMITER ;
root CALL hm_temp_convertor(100, 'F', @res);
Query OK, 0 rows affected, 1 warning (0.00 sec)

rootSELECT @res AS celsius_result;
+----------------+
| celsius_result |
+----------------+
|          37.78 |
+----------------+
1 row in set (0.00 sec)

root
rootCALL hm_temp_convertor(0, 'C', @res);
Query OK, 0 rows affected (0.00 sec)

rootSELECT @res AS fahrenheit_result;
+-------------------+
| fahrenheit_result |
+-------------------+
|             32.00 |
+-------------------+
1 row in set (0.00 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
KD3-98839-Himanshu DELIMITER $$
KD3-98839-HimanshuCREATE PROCEDURE hm_inches_convertor( IN v_inches INT, OUT res VARCHAR(255) )
    -> BEGIN
    -> DECLARE remaining_inches INT;
    -> DECLARE yard INT;
    -> DECLARE foot INT;
    ->
    -> SET  yard = v_inches / 36;
    -> SET  remaining_inches = v_inches % 36;
    ->
    -> SET  foot = remaining_inches / 12;
    -> SET  remaining_inches = remaining_inches % 12;
    ->
    -> SET res = CONCAT(
    ->         yard, ' yard(s), ',
    ->         foot, ' foot/feet, ',
    ->         remaining_inches, ' inch(es)'
    ->     );
    ->
    -> END
    -> $$
Query OK, 0 rows affected (0.02 sec)

KD3-98839-Himanshu DELIMITER ;
KD3-98839-HimanshuCALL hm_inches_convertor(124, @res);
Query OK, 0 rows affected (0.00 sec)

KD3-98839-HimanshuSELECT @res AS result;
+-------------------------------------+
| result                              |
+-------------------------------------+
| 3 yard(s), 1 foot/feet, 4 inch(es) |
+-------------------------------------+
1 row in set (0.00 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
KD3-98839-Himanshu  DELIMITER $$
KD3-98839-HimanshuCREATE PROCEDURE hm_prodct( IN v1 INT, IN v2 INT, OUT res VARCHAR(255) )
    -> BEGIN
    ->
    -> DECLARE product INT DEFAULT 1;
    -> SET  product = v1 * v2 ;
    -> IF product >= 100 THEN
    ->  SET res = 'Product of two no is greater than 100';
    -> ELSE
    ->  SET res = 'Product of two no is less than 100';
    -> END IF;
    -> END
    -> $$
Query OK, 0 rows affected (0.01 sec)
```

```
KD3-98839-Himanshu DELIMITER ;
KD3-98839-HimanshuCALL hm_prodct(5,6, @res);
Query OK, 0 rows affected (0.01 sec)

KD3-98839-HimanshuSELECT @res AS result;
+------------------------------------+
| result                             |
+------------------------------------+
| Product of two no is less than 100 |
+------------------------------------+
1 row in set (0.00 sec)

KD3-98839-HimanshuCALL hm_prodct(25,5, @res);
Query OK, 0 rows affected (0.00 sec)

KD3-98839-HimanshuSELECT @res AS result;
+---------------------------------------+
| result                                |
+---------------------------------------+
| Product of two no is greater than 100 |
+---------------------------------------+
1 row in set (0.00 sec)
```